Univariate Linear Regression

Data Analytics II - Assignment 2

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Introduction

This assignment repeats and sharpens your understanding of univariate linear regression, clarifies the difference between predictive power of models and unbiasedness of coefficient estimators, and lets you estimate univariate linear regression models without the `lm`-function.

You will continue working with a cleaned version of the dataset `ps2_cantons_cleaned.csv` from the exercise session. As reminder, the dataset `ps2_cantons_cleaned.csv` contains the following variables:

Variable	Description
`canton`	**Canton identifier** (Unit of observation)
`gdp`	**Total gross domestic product (GDP) per canton** (in million CHF)
`population`	**Permanent resident population** (in million)
`immigration`	**Immigrant population** (in thousand, including changes in residence status)

In addition, you will add the dataset `ps2_cantons_surnames.csv` on surnames in the cantons of Switzerland. The code from the exercise session might help for this assignment. The new dataset `ps2_cantons_surnames.csv` contains the following variables (Source):

Variable	Description
`canton`	**Canton identifier** (Unit of observation)
`share_mueller`	**Share of the cantonal population with surname "Mueller"/"Müller"** (in percent (100 = 100%))
`share_maier`	**Share of the cantonal population with surname "Maier"** (in percent (100 = 100%)).

Exercises

For grading, sentence and word limits are hard caps: responses that exceed a stated limit lose points.

1. Data preparation [0.5 points]

You want to add a new variable `surname=share_mueller+share_maier` to the previous dataset and make the data ready for regression analysis.

1. Read-in the .csv files. Verify that there are no missing surnames. Create a new variable in `ps2_cantons_surnames.csv` called `surname` which is constructed by adding the shares of `share_mueller` and `share_maier`. Then drop `share_mueller` and `share_maier`.

2. Join the variable `surname` from the dataset `ps2_cantons_surnames.csv` to the dataset `ps2_cantons_cleaned.csv` using the common variable `canton`. Before the join, make sure that canton names coincide. Use tidyverse/dplyr syntax (e.g., `left_join()` with `%>%`).

3. Perform data checks and resolve any issues. Your final dataset must include **all** Swiss cantons, including only the variables `gdp`, `population`, `immigration` and `surname`. The data checks should include

   - no missing surnames after the join
   - there are exactly the 26 unique cantons in the dataset

2. Estimation of an univariate model [1 points]

You want to estimate the effect of `gdp` (as Y) on `surname` (as X) incl. an intercept.

1. State the regression equation of *this specific* univariate linear model, including the error term. Don't write the generic univariate linear regression formula.

2. Estimate the model coefficients using the `lm`-function. Report coefficients and interpret them. Use no more than 2 sentences *per* coefficient.

3. Use the estimated regression model to predict the value of `gdp` based on the average share of Mueller and Maier in the full Swiss population. Explain in 1 sentence the predicted change in `gdp`.

3. Comparison of two univariate models [1.5 points]

You want to compare the estimated model from subexercise 2 "Estimation of an univariate model" to the estimated regression of `gdp` (as Y) on the natural logarithm of `immigration` (as X) incl. a constant.

1. Run the corresponding regression.

2. Report and compare the R^2 of the two regressions, and comment on their predictive power. Answer the following questions:

   - What does R^2 measure here intuitively? (max. 2 sentences for your answer)
   - What amount of variation in `gdp` is explained? (max. 2 sentences for your answer)

3. Discuss with max. 200 words whether the estimators for the coefficients on `surname` and `immigration` are unbiased in these two models (Consider SLR.4 and potential violations, e.g. omitted variable bias).

4. In general, how are (i) the unbiasedness of a coefficient estimator and (ii) the predictive power of a model (e.g., measured by R^2) related? Discuss with max. 3 sentences.

4. Univariate model without a constant [0.5 points]

The goal is to regress `gdp` (as Y) on `surname` (as X) *without* a constant.

1. State the formula of *this specific* univariate linear model.

2. Estimate the model coefficient using the `lm`-function and report it. Check the function documentation to find out how to remove the constant.

3. Why did the coefficient on `surname` change compared to subexercise 2.? Discuss with max. 2 sentences.

5. Code your own OLS estimator [1.5 points]

1. Write your own R-function to estimate the parameters of a univariate regression model, ***without using the `lm()` function or any other pre-existing function from a package that has an implementation of a linear regression model***. The goal is that you implement the estimation of the parameters yourself.

   - Your function must take the vectors `y` and `x` as inputs and return the estimated coefficients from an univariate regression model.

- You only need to estimate the coefficients, not their standard errors.
- Make sure to have sufficient in-line documentation for someone else to understand your code.

2. Verify your function against the `lm()` function using the regression from subexercise 2.
3. As a last step, modify your function by adding the boolean input `add_constant` to the function. Use an `if` condition to implement the estimation procedure for the case without a constant and return only one coefficient in this case. Verify as before by comparing against your results in subexample 3.